A Multi-Agent Framework for Multicultural Text-to-Image Generation

PROJECT REPORT [Phase-I]

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CERTIFICATE

This is to certify that the project report entitled "**A Multi-Agent Framework for Multicultural Text-to-Image Generation**" was completed to our satisfaction by **[Ayush Patel (24U020046)], [Gaurav Kumar (24U020036)]**. The work is a bonafide piece of research and development carried out under our guidance in the **Department of Computer Science and Engineering, Indian Institute of Information Technology Bhopal**, in partial fulfillment of the requirements for the degree of Bachelor of Technology during the academic year [2026].

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DECLARATION

We declare that the project report entitled "**A Multi-Agent Framework for Multicultural Text-to-Image Generation**" submitted in partial fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of our own work, carried out under the guidance of **Dr. Ringki Das**. The work was done entirely at the Indian Institute of Information Technology Bhopal, and it has not been submitted, wholly or in part, for any degree or diploma at any other institute or organization.

All ideas, text, data and figures used from other sources have been acknowledged in the reference section. All information given is accurate to the best of our knowledge. We will take responsibility if any inaccuracies are found.

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AREA OF WORK

This project is in the field of Generative AI and Multimodal Machine Learning. We used several technologies, including text-to-image diffusion models and large language models, to create collaborative agents that generate text and to automatically and manually assess the generated images. We were interested in seeing what impact culture would have on the text and, therefore, on the accuracy of the generated image. If two cultures are involved in the request (for example, someone of one nationality visiting a landmark in another country), the quality of the text affects the accuracy of the image. We have developed a captioning pipeline to test distributing the writing of the text among multiple role-playing agents.

A robust system was interesting to us. We made thousands of calls to both language models and image generators. As we were using shared hardware, we put a lot of effort into making sure our data was stored and retrieved robustly. This meant we could stop and start our run at any time.

We also have an interactive studio which uses the pipeline to serve regular requests.

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ABSTRACT

Text-to-image models are regularly asked for scenes in which a person from one culture stands in a place that belongs to another, yet the caption handed to the model is usually a single templated sentence. This project studies whether dividing the writing of that caption among several role-playing language-model agents keeps the generated picture closer to the requested nationality, age, gender and landmark. Using the task definition of Bhalerao et al. [1], we built Multi-Agent Framework, an independent end-to-end implementation that pairs a three-agent, two-round caption composer, served by Llama-family models through the Groq API, with AltDiffusion-m18, SDXL and, when hardware permits, FLUX.1-lite as image generators.

Our prompt grid holds 750 person–landmark combinations (five nationalities, three age groups, two genders and 25 landmarks), which grow to 3,750 prompts across five languages. Five automatic measures are computed for every image: CLIP alignment, Inception Score, a SigLIP-based aesthetic score, a landmark-swap knowledge probe and a demographic-substitution fairness probe. The principal engineering contribution is fault tolerance. Every caption, image and score is written to persistent storage the moment it exists, images are saved through a temporary file and renamed, a caption produced by the template fallback is never cached, and a repeated run resumes from the first missing item. This allowed a benchmark to survive repeated disconnections of a free-tier GPU runtime. We add a blinded human-rating pack with weighted Cohen's kappa and a Gradio studio that generates several candidates for a request, ranks them with a weighted blend of alignment, landmark knowledge and aesthetics, and returns the best one.

This report explains the system design and the experiments that are set up in the code. It does not claim that local experiments have been completed. Numerical results from the research paper are labelled as published results. A full run and human review are needed before conclusions can be made about the local implementation.

Keywords: multicultural images, multi-agent system, text-to-image generation, diffusion models, image evaluation, reproducibility.

1. INTRODUCTION

1.1 Background

Whilst appearing to be a straightforward prompt, generating an image from the prompt "A Vietnamese girl in front of the Golden Gate Bridge" requires the image generator to be aware of two cultures. The image will include cultural aspects such as clothing, features and body language from one culture and architecture and environment from another. Latent diffusion models [2], [3] have been created to generate realistic images from very short text prompts, and are commonly used in systems such as SDXL, DALL·E 3 and FLUX [4], [5], [6]. However, the captions used to train these models are biased towards a small number of languages, and focus on adult, western cultures and males [7], [8]. This can lead to stereotypical images, incorrect architecture and half of the prompt being ignored.

Firstly, the majority of research into testing cultural competence in image generation considers only one culture within the image (for example, where the food, festival or monument belongs to a single origin) [9], [10], [11]. Secondly, the caption is often the weakest part of the prompt. As it is created as one line of text, it carries no information about the materials, colours, lighting and architectural aspects of the clothing and scene. The image generator therefore fills in all the gaps from its own prior knowledge.

Recently, the term multicultural text-to-image generation has been coined by Bhalerao et al. [1]. They suggest that a caption should be created from a dialogue between language-model agents, each with a persona: one represents the nationality mentioned, one is an informed tourist of the landmark, and one can describe the age and gender of the subject. This project follows that definition of the task and focuses on making the method repeatable, reliable and practical.

1.2 Motivation

The multimodal benchmark is quite intensive to run on lower-end hardware. It requires many calls to the language model and the diffusion models. The free runtime of a notebook can cut out midway. There are also issues with rate limiting on hosted language models. In the worst case, a failed call silently substitutes the template caption, which means the multi-agent method is actually being compared with a hidden copy of the baseline.

There are also issues with the usability of the results. Most research code returns a table of averages. If someone just wants a nice picture of their grandmother from one country visiting a landmark in another country, they cannot use those results. We wanted to create an interactive tool, using the same captioning system, that would give a few options and return the best match to the request.

1.3 Scope of the Work

This project includes the design of the system and the experiments. There was no training or fine-tuning of any image model. The generators are publicly available, and the language model is an open-weight deployment. Captions were created, experiments were made robust to interruptions, and results were quantified and visualised. The demographics are limited (5 nationalities, 3 age groups, 2 genders). All automatic metrics are proxies and should be treated with caution. Cultural authenticity is subjective, and a protocol has been provided for human scoring.

1.4 Contributions

- A completely self-contained multi-agent system consisting of a moderator, three persona agents, a summarization agent and a deterministic length controller (the caption governor) that keeps every caption within the 77-token limit of the CLIP text encoder.
- A fault-tolerant benchmarking system that saves captions, images and scores for each item. The benchmark can be restarted from any point and never stores a low-quality fallback caption.
- A landmark-swapping knowledge probe and a demographic-swapping fairness probe, with results broken down by age, gender, person country, landmark country and caption language.
- A blind human-evaluation system with rating sheets containing hashed sample IDs. Completed ratings cannot be changed. Weighted Cohen's kappa [12] and error tags are provided.
- An interactive studio that takes a free-text query, generates a caption and shows a number of candidate images, ordered by a weighted sum of three normalised signals. Design decisions are explained, together with what can and cannot be concluded from the automatic scores.

1.5 Organization of the Report

Chapter 2 gives an overview of the literature and shows the opportunity that the project looks to address. Chapter 3 gives the problem statement and the project aims. Chapter 4 gives details of the methodology. Chapter 5 gives details of the algorithms and their costs. Chapter 6 gives details of the block diagrams, flowcharts and data flow diagram. Chapter 7 gives details of the tools used and the implementation. Chapter 8 gives details of the evaluation method, results and discussion. Chapter 9 gives the conclusion and future work. Chapter 10 gives the list of references.

2. LITERATURE REVIEW

This chapter surveys the strands of research on which the project rests: diffusion-based image synthesis, multilingual generation, evaluation of text–image agreement, cultural competence and bias, and multi-agent prompting with language models. It closes with a comparison table and a statement of the research gap.

2.1 Diffusion-Based Text-to-Image Synthesis

Denoising diffusion probabilistic models are trained to reverse the process of gradually adding noise. Once trained, they can generate samples by reversing a large number of small noising steps [2]. The cost of denoising was reduced by working in the compressed latent space of an autoencoder rather than in pixels, which allowed text-conditioned generation on a single GPU [3]. The denoising network was later enlarged, a second text encoder was added, and the model was conditioned on image size and crop to help composition at higher resolutions [4]. A large part of the improvement in DALL·E 3 came from training on richer machine-generated captions [5]. This is important: if richer captions help during training, they should also help during inference. FLUX.1 uses a large transformer as its backbone [6]. Distilled models sacrifice some fidelity to gain a smaller memory footprint.

2.2 Multilingual Generation

Most generators accept English only. AltDiffusion replaces the CLIP text encoder with a multilingual one and trains in two stages: a distillation stage that aligns the new encoder with the original, and a contrastive fine-tuning stage on multilingual pairs, so that prompts in a number of languages steer the same image decoder [13]. It is one of very few open models that let us ask for the same scene in English, German, Hindi, Spanish and Vietnamese, which is why it is the default generator in this project. The price is a 77-token limit on the text side, a constraint that shapes our caption governor in Chapter 4.

2.3 Measuring Text–Image Agreement and Image Quality

CLIP was trained to learn a shared embedding space between images and text using hundreds of millions of image–text pairs from the web [14]. CLIPScore measures image–caption agreement as the cosine similarity between the image and text embeddings [15]. It is easily calculated for all images in a large benchmark; however, its lack of granularity can give high scores to images that contain the correct objects in the wrong arrangement. The Inception Score measures the sharpness and diversity of a collection of images from the class predictions made by an Inception network [16], [17], but it does not consider the relevance of the content. An aesthetic predictor built on a pretrained image–text encoder with a sigmoid loss produces a score between 1 and 10 for the aesthetic quality of an image [18], but it does not consider whether the image is semantically correct.

Larger benchmarks use a variety of approaches. HEIM scores models on alignment, quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality and efficiency [19]. TIFA uses a question-answering model to ask questions about the generated image and so determine faithfulness [20]. GenEval uses object detectors to check the number, colour and position of objects [21]. None of these methods takes two cultures in one scene into account, and none is aimed at determining whether an image is stereotypical or accurate.

2.4 Cultural Competence in Language and Vision Models

Surveys have shown that defining cultural aspects of language technologies is challenging, as culture is often treated as synonymous with country and language [22], [23]. Large language models have been shown to be biased towards Western defaults even when prompted in a different language [24]. In the visual domain, CVQA collects culturally grounded questions in many languages [9], GlobalRG evaluates retrieval and grounding of culture-specific concepts [10], and CUBE probes the cultural competence of text-to-image models through the generation and diversity of culturally specific items [11]. A study of the cultural gap in text-to-image generation reports that models perform better for high-resource cultures and languages, and that bilingual or multilingual encoders narrow the gap only partly [7]. Work on demographic stereotypes shows that freely available generators amplify them at scale, for example by returning narrow depictions of occupations or nationalities [8].

2.5 Multi-Agent Prompting with Language Models

Open-weight language models of the Llama family [25] can adopt a persona from a short system message and answer in character. Surveys of multi-agent systems describe how several such agents, each with its own role, can critique and extend one another's answers [26]. Bai et al. applied this idea to cultural image captioning, where agents with different cultural backgrounds discuss an image and produce a fuller description than a single model does [27]. That work describes existing images. Generation inverts the direction: the description must be written first, and the picture is judged against it.

2.6 The Reference Study

The reference study [1] provides a benchmark dataset of 9,000 images spanning 5 countries, 3 age groups, 2 genders, 25 landmarks and 5 languages. It also compares how well simple prompts work against richer prompts created by a moderator, three persona agents and a summarizer. The main results showed that image quality, landmark knowledge and consistency improved with the richer prompt, while alignment and aesthetics did not improve. It was also found that alignment was better for English prompts. The size of the grid and the five factors we measure have been taken from this study, and its published numbers are used in Chapter 8 to give context.

2.7 Comparative Summary

Table 1: Comparison of related work with the present project

Work	Main contribution	Cultural setting	Limitation for our purpose
Latent diffusion [3]	Diffusion in autoencoder latent space	None	No cultural analysis
SDXL [4]	Larger U-Net, dual text encoders	None	English prompts; no cultural probe
DALL·E 3 [5]	Training on synthetic detailed captions	None	Closed model; captions not exposed at inference
AltDiffusion [13]	Multilingual text encoder for diffusion	Language coverage	77-token limit; no person–landmark study
CLIPScore [15]	Reference-free alignment metric	None	Coarse; weak on composition
HEIM [19]	Holistic multi-aspect benchmark	General social aspects	One culture per image at best
TIFA / GenEval [20], [21]	QA- and detector-based faithfulness	None	Not designed for cultural cues
CVQA / GlobalRG [9], [10]	Culturally grounded VQA and retrieval	Many cultures, one per item	Understanding, not generation
CUBE [11]	Cultural competence of T2I models	Single culture per image	No cross-cultural scenes
Cultural gap study [7]	Quantifies gap across cultures and languages	Single culture	No caption-composition remedy
Stereotype audit [8]	Shows demographic amplification	Demographic	Diagnoses but does not mitigate
Multi-agent captioning [27]	Agents caption existing images	Single culture	Image-to-text direction
Reference study [1]	Benchmark and multi-agent prompt composition	Person plus landmark, five languages	Research code; no resumable runner or interactive tool
This project	Resumable pipeline, probes, human pack, ranking studio	Person plus landmark, five languages	Small sample; proxy metrics

2.8 Critical Discussion

From Table 1, it can be seen that the related work falls into three distinct groups, with very little overlap. The generator papers [4], [5], [6], [13] each improve some aspect of fidelity or language, but are scored using generic prompts. The measurement papers [19], [20], [21] improve the measurement of text–image agreement, but rarely include a human at a specific cultural place in their prompts. The cultural papers [7], [9], [10], [11] ask appropriate questions, but mostly focus on understanding, or on generation for a single culture.

Methodologically, any metric based on CLIP will inherit the bias of CLIP's training data [14], so scores could vary between demographic groups depending on the evaluator. The fairness and knowledge probes are therefore treated as strategies for comparing conditions against one another with the same evaluator. It can also be difficult to know which improvements come simply from having a longer prompt; for example, DALL·E 3 benefits from added detail [5]. To measure multi-agent composition fairly, a single-agent caption of the same length should also be included. This is noted as future work, as we used the design of the original study [1] in our implementation.

Whilst the prompts, models and metrics are described in these papers, details of how to run thousands of generations on hardware that can be lost are rarely given. This is an important point for reproducibility, so the checkpointing design is part of our contributions. Finally, whilst nationality, age and gender are common demographic axes in the literature, they are a simplification [22], [23]. This is discussed further when considering the validity of our conclusions.

2.9 Research Gap

We identified three issues. Firstly, most benchmarks and models assume that each image contains only one culture, and there is little research into cross-cultural scenes. Secondly, whilst multi-agent captioning has been shown to be useful, the available code does not account for the disruptions that occur on shared or free hardware. Thirdly, automatic scoring is a proxy measure with limitations, and there is generally no clear protocol for human scoring. This project tackles the second and third issues directly, and the first through the design of its scoring.

3. PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

Given a request with details about the nationality, age group and gender of a person, a landmark (potentially from a different country) and a language for the caption, generate an image in which the person and the landmark are clearly visible and the image is culturally appropriate.

Formally, given a request $r = (n, a, g, l, \lambda)$, where n is the nationality, a the age group, g the gender, l the landmark and λ the language of the caption, a captioning schema σ takes the request r and produces a caption $c = \sigma(r)$ of at most 77 CLIP tokens. A generator function G takes a caption c and a seed s and produces an image $I = G(c, s)$.

Our task has three parts. First, design a captioning schema σ that gives better alignment between the generated image and the request r than a simple template. Second, quantify the alignment, landmark knowledge and demographic-substitution robustness in a repeatable way with minimal hardware. Third, demonstrate this as a tool that can produce the best image from multiple options.

3.2 Challenges

Under-specified prompts: The prompts are not fully specified, so the generator has to make assumptions about the clothes, materials, colours and context. Only the nationality and landmark information is given.

Text length: CLIP's text encoder can only take 77 tokens, so the caption must be short, and the name of the landmark must still be included in it.

Unreliable services: The language models are online services and have rate limits. Free GPU sessions may also drop out.

Poor proxy measures: The alignment, quality and aesthetic scores do not indicate how culturally correct an image is. Two of these scores can be increased when the wrong monument is shown.

Fairness: The baseline and the multi-agent method must be run with the same seed, prompts and settings. If a caption has been cut down to a fallback, it must not be counted as a multi-agent result.

3.3 Objectives

1. Implement a moderator, three persona agents and a summarizer, in a two-turn dialogue that generates a caption (maximum 48 words, 77 tokens).
2. Implement a benchmark runner that uses the 750-prompt grid (expandable to five languages), stores all intermediate results, and can be paused and continued.

3. Implement five automatic metrics for alignment, quality, aesthetics, knowledge and fairness across demographics and cultural factors.
4. Provide a human-evaluation package with inter-rater statistics, using a fixed set of error tags.
5. Provide a studio interface for converting free-text requests into a ranked list of image candidates, with all runs logged with metadata.
6. Clarify what the metrics can and cannot measure.

3.4 Success Criteria

1. A moderator, three persona agents and a summarizer are created, and a caption of up to 48 words (77 tokens) is produced via a two-turn dialogue.
2. The benchmark has been completed. No duplicate captions were created on repeat runs. All captions were within the 77-token limit.
3. The studio returns an ordered gallery, and all requests are logged.
4. There is a clear difference between the captions created by the simple strategy and those created by the multi-agent strategy.

4. PROPOSED METHODOLOGY AND WORK DESCRIPTION

The work is organised as a sequence of stages that share a persistent store. Fig. 1 in Chapter 6 gives the overall block diagram; this chapter explains each stage and the reasons behind the design choices.

4.1 Prompt Grid

The grid crosses five nationalities with three age groups, two genders and 25 landmarks, five landmarks for each country. That gives 5 × 3 × 2 × 25 = 750 base prompts. Because each nationality is matched with five landmarks from its own country and twenty from the other four, 150 prompts place a person at a landmark of their own culture and 600 are cross-cultural. Crossing the base prompts with five caption languages yields 3,750 prompts. Table 2 lists the dimensions.

Table 2: Dimensions of the prompt grid

Dimension	Values	Members
Nationality	5	German, Indian, Spanish, American, Vietnamese
Age group	3	Child, adult, elder
Gender	2	Female, male
Landmark	25	Five per country, for example Cologne Cathedral and Brandenburg Gate (Germany), Taj Mahal and Charminar (India), Alhambra and Sagrada Familia (Spain), Statue of Liberty and White House (U.S.), One Pillar Pagoda and Independence Palace (Vietnam)
Caption language	5	English, German, Hindi, Spanish, Vietnamese
Base prompts	750	5 × 3 × 2 × 25; 150 same-culture and 600 cross-cultural
Expanded prompts	3,750	750 × 5 languages

Age and gender are mapped to short descriptive phrases (for instance a twelve-year-old girl, a ten-year-old boy, a thirty-five-year-old woman or man, and an elderly woman or man) so that the agents and the translator receive the same wording.

4.2 Caption Strategies

4.2.1 Simple caption

The baseline is a fixed sentence: a nationality adjective, a noun for the person, the phrase “wearing traditional attire” and the landmark. For a Vietnamese girl at the Golden Gate Bridge it reads “A Vietnamese girl wearing traditional attire, standing in front of the Golden Gate Bridge.” The baseline is deterministic, so it costs nothing and can never fail.

4.2.2 Multi-agent caption

The multi-agent strategy uses five roles (Table 3). The moderator is an ordinary function that packs the request into three task descriptions. Three persona agents, played by one hosted language model under different system messages, describe the visible aspects of the scene: the clothing of a person of the given nationality, the architecture and surroundings of the landmark, and the physical traits of a person of the given age and gender. In the first round, each agent writes a description of fewer than 25 words. In the second round, each agent sees what the others wrote and is asked a pointed question that forces interaction, for example how the clothing would harmonize with the colours of the landmark, or what the age–gender description should change so that the attire suits the person. After two rounds, a summarizer agent merges the three descriptions into one prompt that must mention the nationality, the age and gender, and the landmark.

Table 3: Agent roles and responsibilities

Agent	Persona	Task
Moderator	Plain function, no model call	Receives nationality, age, gender and landmark, and assigns focused tasks to the persona agents
Country agent	A person from the given country who knows its culture	Describes clothing, accessories, materials, patterns and symbolic colours
Landmark agent	A frequent visitor of the landmark	Describes architecture, colours and environment
Age–gender agent	A person of the given age group and gender	Describes attire, accessories, skin, hair texture and body traits
Summarizer	A writer of concise visual prompts	Fuses the three descriptions into one line of fewer than 48 words and 77 tokens

One multi-agent caption therefore costs 3 + 3 + 1 = 7 language-model calls with two rounds. For the full grid this means 5,250 calls for English captions alone, a figure that explains why rate limits and caching matter.

4.2.3 Caption governor

Language models cannot give an accurate CLIP token count, yet the CLIP text encoder accepts only 77 tokens. The summarizer's answer is therefore passed through a caption governor. The governor keeps only the first line, strips any unneeded characters, and limits the caption to 48 words. If the name of the landmark is not included in the caption, the phrase "in front of the" followed by the landmark is added. The caption is then tokenized with the CLIP tokenizer. If it is over 74 tokens, the last comma-separated phrase that does not contain the landmark is removed, repeatedly. If nothing more can be removed this way, the caption is shortened by removing words from the end. The 74-token limit leaves 3 tokens of the 77-token window as a buffer for the start and end tokens. See Fig. 4.

4.2.4 Failure policy

If the first calls get no response, the moderator supplies a default caption containing some neutral information about the image, so that the studio can still be used. We must make sure this is never mistaken for a multi-agent result: the conversation log states that a default caption was used. The benchmark runner refuses to cache such a caption and raises an error instead, but everything saved so far remains on disk. When the benchmark is run again, only the missing items are computed.

4.3 Translation

Captions are always composed in English. For the other four languages, the English caption is translated by the same language-model service and cached under a key that includes the language. Scoring is always done against the English caption, so metrics remain comparable across languages. When a model cannot read the target language (SDXL and FLUX accept English only), the English caption is used directly.

4.4 Image Generation

Three generators are supported (Table 4). AltDiffusion-m18 reads multilingual text and runs on a 16 GB T4. SDXL-base is a fast English-only alternative. FLUX.1-lite (8B) follows the reference study but needs about 24 GB of GPU memory and 25 GB of system memory, which a free runtime does not offer. Each model has a “paper” preset with high resolution and many steps, and a “fast” preset for experimentation. Both presets fix the seed, and the seed for case i is the base seed plus i, so the simple and multi-agent images of one case start from the same noise and differ only through the caption.

Table 4: Generator presets used in the implementation

Model	Preset	Resolution (px)	Steps	Guidance	Seed
AltDiffusion-m18	paper	768	110	11.0	11000
AltDiffusion-m18	fast	512	30	8.0	11000
FLUX.1-lite 8B	paper	768	30	4.0	11
FLUX.1-lite 8B	fast	512	20	3.5	11
SDXL-base 1.0	paper	768	30	6.5	11
SDXL-base 1.0	fast	512	20	6.5	11

4.5 Studio Ranking

The studio accepts unstructured or structured input. A keyword parser extracts the nationality, age, gender, landmark and language, and a limited model call is used to ensure that only valid answers are returned. If the nationality and landmark have been identified, the chosen strategy is used to create the caption. Otherwise, one sensible line is created in English from all the information given by the user, without adding stereotypes or extra people. N candidates are

then generated with seeds s, s+1, …, s+N−1, and each candidate is scored on the following blend of signals:

$$final = (0.45 \cdot mm(alignment) + 0.30 \cdot mm(knowledge) + 0.25 \cdot mm(aesthetic)) / (\text{sum of the weights of the available signals})$$

where mm(·) is min–max normalisation within the batch. If a candidate has no value for a signal, 0.5 is used. If no candidate in the batch has a value for a signal, that signal is dropped and the sum of the weights of the remaining signals is used. If there is only one candidate, every signal is given 0.5, so ranking is only meaningful when $N \geq 2$. The best image and a metadata file with the parsed fields, captions, seeds, scores and agent log are saved in a timestamped folder.

4.6 Evaluation Measures

Five automatic measures are computed for every image (Table 5). Their implementation is described in Section 7.7.

Table 5: Evaluation measures and their definitions

Measure	Definition	Better
Alignment	CLIP ViT-L/14 similarity between the image and the English caption: the dot product of the normalised image and text embeddings [14], [15]	Higher
Quality	Inception Score from Inception-v3 class predictions on images resized to 299 × 299, averaged over five splits [16], [17]	Higher
Aesthetics	SigLIP-based aesthetic predictor score on a 1–10 scale [18]	Higher
Knowledge	Landmark-swap probe: the CLIP similarity to the original caption minus the similarity to the caption in which the landmark is replaced by one from a different country	Higher
Fairness	Demographic-substitution probe: sensitivity of the result when the gender, age group or nationality of the request is substituted	Lower

5. PROPOSED ALGORITHMS

Four procedures carry the method. Algorithm 1 composes a multi-agent caption, Algorithm 2 enforces the token budget, Algorithm 3 runs a resumable benchmark and Algorithm 4 ranks candidates in the studio. The notation follows Chapter 3: n, a, g, l and λ are nationality, age group, gender, landmark and language, R is the number of rounds and A the number of persona agents.

Algorithm 1: Multi-agent caption composition

```
Input: n, a, g, l, rounds R (default 2)
Output: caption c, conversation log L
1  tasks ← Moderator(n, a, g, l); L ← [tasks]
2  for each agent X in {Country, Landmark, AgeGender} do
3      d[X] ← ask(persona(X), task(X), limit 25 words)
4  if any d[X] is empty then return Fallback(n, a, g, l),
       L marked as fallback
5  for r ← 2 to R do
6      for each agent X do
7          d'[X] ← ask(persona(X), d[X], peer descriptions,
                       probing question)
8      for each agent X do d[X] ← d'[X] if non-empty else d[X]
9  s ← ask(Summarizer, d[Country], d[Landmark], d[AgeGender])
10 if s is empty then return Fallback(n, a, g, l),
       L marked as fallback
11 c ← Governor(s, n, l)
12 return c, L
```

Algorithm 2: Caption governor (token budget)

```
Input: text s, landmark l, word limit W = 48, token limit T = 74
Output: caption c
1  t ← first line of s without quotes or leading labels
2  if |words(t)| > W then t ← first W words of t
3  if l does not occur in t then t ← t + ", in front of the " + l
4  P ← list of comma-separated phrases of t
5  while tokens(join(P)) > T and |P| > 1 do
6      find the last index i > 1 with l not in P[i]
7      if none exists then break else remove P[i]
8  c ← join(P)
9  while tokens(c) > T and |words(c)| > 10 do remove the last word of c
10 return c with a final full stop
```

Algorithm 3: Resumable benchmark

```
Input: model M, case count N, strategies S, languages Λ, seed s0
Output: table of per-image metrics
1  cases ← first N rows of the grid shuffled with seed s0
2  for each case i, strategy σ, language λ do
3      key ← (σ, n, a, g, l)
4      if key not in cache then cache[key] ← Caption(σ, …); save cache
5      if λ ≠ English and (key, λ) not in cache then translate; save cache
6  for each row whose image file is missing or unreadable do
7      I ← Generate(M, caption, seed s0 + i); write temp file; rename
8  for each row whose score key is missing in the score log do
9      compute CLIP, knowledge and aesthetic scores; append one line; flush
10 write the per-image table; return it
```

Algorithm 4: Best-of-N ranking in the studio

```
Input: request text or fields, strategy, N, seed s, weights w = (0.45, 0.30, 0.25)
Output: best image and ranked table
1  fields ← Parse(request); caption c ← Strategy(fields)
2  for j ← 0 to N − 1 do I[j] ← Generate(c, s + j); save I[j]
3  for each j do alignment[j] ← S(c, I[j]);
                knowledge[j] ← S(c, I[j]) − S(c′, I[j])
4  aesthetic[j] ← AestheticScore(I[j])
5  for each available signal z do z′ ← minmax(z)
       (0.5 if constant or missing)
6  final[j] ← Σ w_z · z′[j] / Σ w_z over available signals
7  sort candidates by final; save BEST and metadata; return
```

5.1 Complexity

Given K phrases and L tokens in a line from the summarizer, Algorithm 1 requires $A \cdot R + 1$ model calls, which for $A = 3$ and $R = 2$ is 7 calls, regardless of grid size. Algorithm 2 re-tokenises after each deletion, so it has a time complexity of $O(K \cdot L)$ and a space complexity of $O(L)$. In practice K is much less than 10, so the governor is negligible compared with the network call.

Algorithm 3 requires at most $|\text{cases}| \cdot |S| \cdot |\Lambda|$ caption look-ups, which is also the maximum number of image generations and of scoring passes. The dominant factor is image generation. An image requires a number of denoising steps proportional to the step parameter of the preset, so moving from the fast AltDiffusion preset (30 steps) to the one used in the reference paper (110 steps) increases the amount of denoising by a factor of about 3.7. Reading the score log at start-up is linear in its length. Algorithm 4 needs N generations, N CLIP passes for alignment, N further text passes for the knowledge probe and N aesthetic passes, so time grows linearly with N, and memory stays constant because candidates are written to disk as they appear.

For the fairness probe, one base case produces an original and three substitutions (gender, age, nationality) per strategy. With two strategies that is eight images per base case, so four base cases give 32 images. The default benchmark of twelve cases with two strategies in English produces 24 images, and the human-rating pack defaults to 24 images.

6. PROPOSED FLOWCHART, BLOCK DIAGRAM AND DFD

6.1 System Block Diagram

The prompt grid acts as input to the caption builder (Fig. 1). The caption builder can produce simple and multi-agent captions. These captions are fed into the translation layer and then to the generator. The generated images receive scores from the evaluator bank and the two probes. All of this information is also written to the checkpoint store, and the data is used in the human-rating pack and the analysis plots. The studio uses the same caption builder, generator and evaluators.

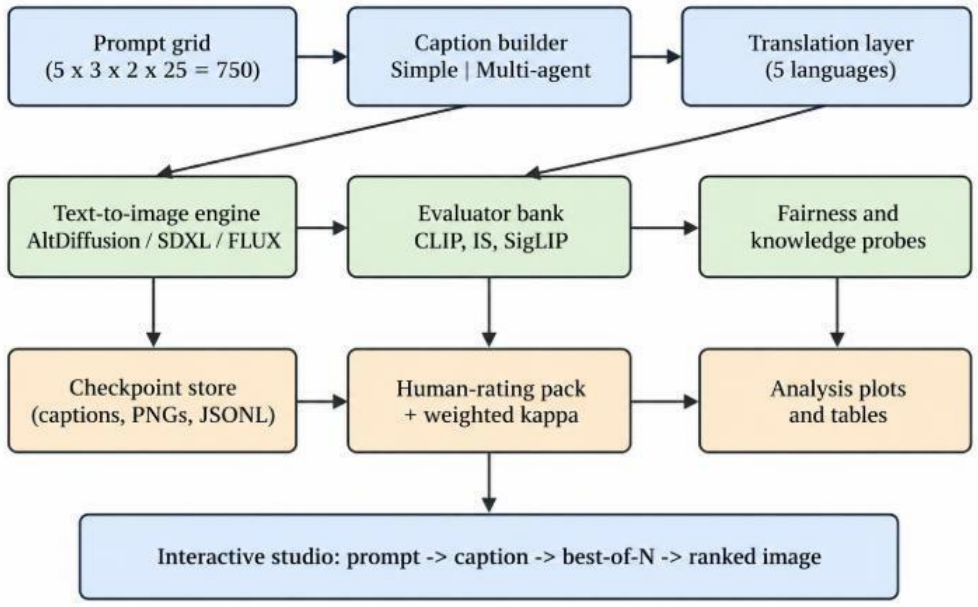


Fig. 1: Block diagram of the Multi-Agent Framework

6.2 Agent Conversation

Fig. 2 shows the interaction between the agents. The moderator splits the request into three tasks. In the first stage, each persona agent creates a short description. In the second stage, each creates a more detailed description based on its peers' descriptions. The summarizer receives only the detailed descriptions.

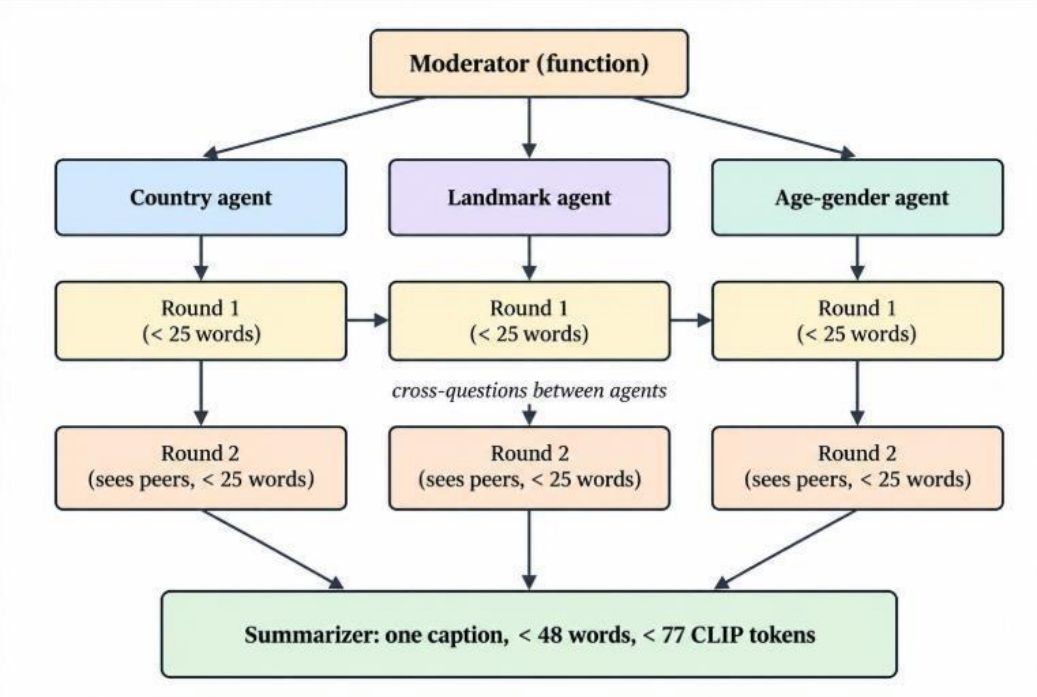


Fig. 2: Two-round conversation among the persona agents and the summarizer

6.3 Flowchart of the Resumable Runner

A standard run is shown in Fig. 3. The runner uses cached data, re-uses captions from previous runs, generates images only where none exist or where the existing ones are unreadable, and scores only the rows that have not yet been scored. If the run is stopped, it can simply be repeated.

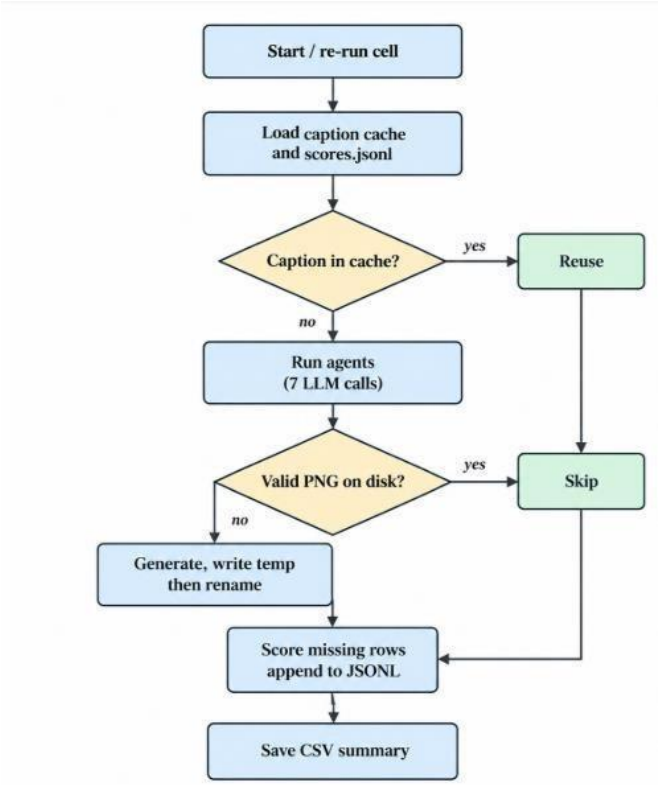


Fig. 3: Flowchart of a resumable benchmark run

6.4 Flowchart of the Caption Governor

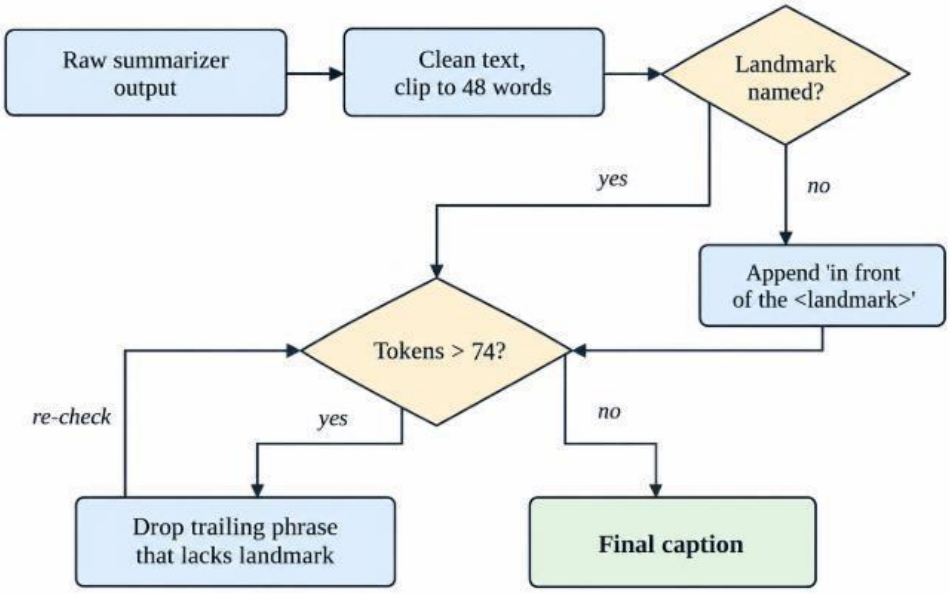


Fig. 4: Flowchart of the caption governor that enforces the 77-token window

The governor never removes the phrase that names the landmark, because the landmark is the element on which the knowledge probe depends. Phrases that describe light or mood are the first to go.

6.5 Data Flow Diagram

Fig. 5 is a level-1 data-flow diagram of the studio. A request enters process 1.0, which extracts fields. Process 2.0 composes the caption, consulting the language-model service and the caption cache. Process 3.0 renders candidates, and process 4.0 scores and ranks them. Images and scores are kept in store D2, from which the rating sheet for human raters is drawn.

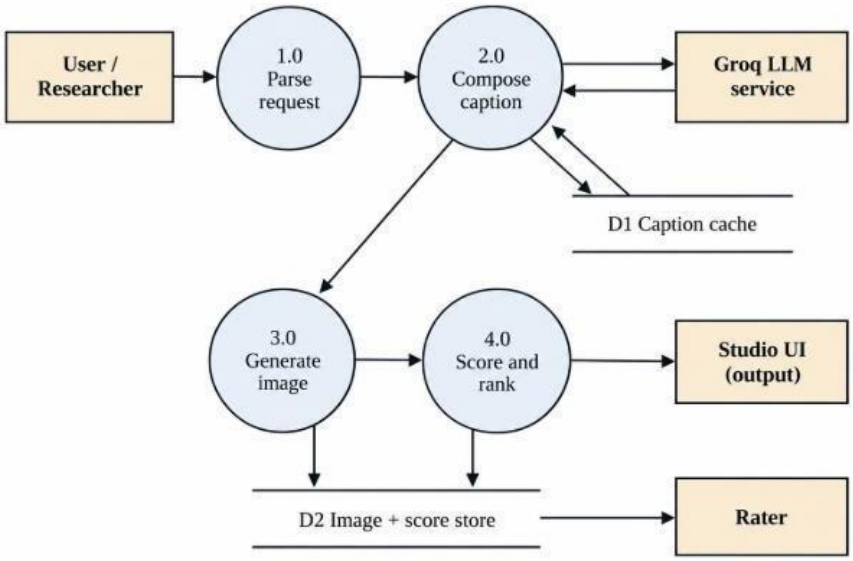


Fig. 5: Level-1 data flow diagram of the studio and benchmark

6.6 Ranking Block Diagram

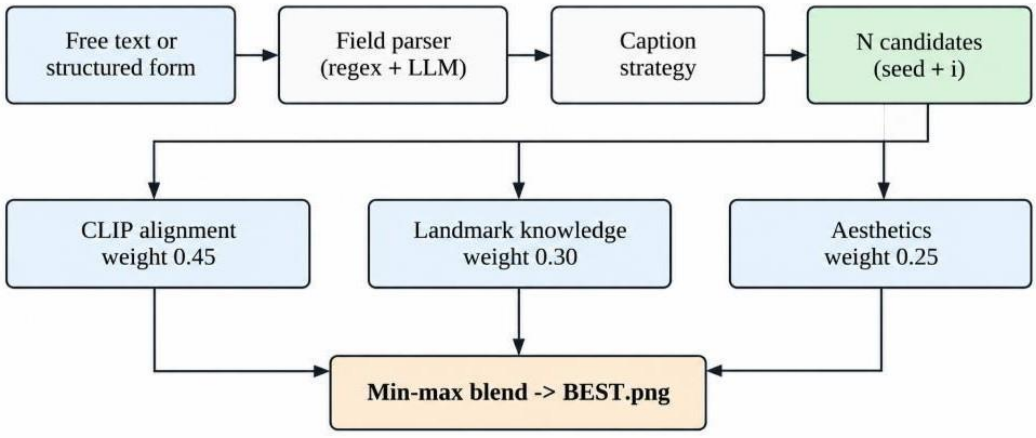


Fig. 6: Block diagram of best-of-N candidate ranking

The weighting reflects the priority given to the three criteria: alignment (0.45), landmark knowledge (0.30) and visual appeal (0.25). The whole purpose of the studio is to produce an image that shows what the user asked for, so agreement with the request carries the most weight.

7. IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

7.1 Execution Environment

All of this work was done in a hosted Python notebook with one NVIDIA T4 GPU. Google Drive is mounted so that, if the runtime crashes, the captions, images, scores, rating sheets and studio outputs are saved. The model weights (about 10 GB) are not saved on Drive because there is not much space, so each time a new runtime is created the weights have to be downloaded again, which takes a few minutes. If Google Drive cannot be mounted, the notebook saves locally and prints a warning that everything will be lost when the runtime disconnects. A flag prevents images from being created if the runtime is using a CPU (where an image can take minutes), without affecting any previous work.

7.2 Software Stack

The package versions used by the generators are fixed (Table 6), as the diffusion pipelines change between versions; the PyTorch package is not modified. The Hugging Face libraries provide model loading and the evaluation functions for CLIP and SigLIP, and Gradio provides the interface.

Table 6: Software components and their roles

Component	Version	Role
Python notebook runtime	T4 GPU session	Execution of all stages
PyTorch [28]	As provided	Tensor computation, Inception-v3 weights via torchvision
diffusers	0.35.1	AltDiffusion, SDXL and FLUX pipelines
transformers [29]	4.55.4	CLIP ViT-L/14, CLIP tokenizer, text encoder classes
accelerate / huggingface_hub	1.10.1 / 0.34.4	Device placement, resumable snapshot download
Groq Python client	Latest	Hosted Llama-family language models for agents and translation
Gradio [30]	5.49.1	Browser interface of the studio
pandas, scikit-learn, matplotlib	Latest	Tables, weighted kappa, plots
aesthetic-predictor-v2-5	Latest	SigLIP-based aesthetic score

7.3 Language-Model Service Layer

The agents use the Groq API. The notebook cycles through a list of models and uses the first one that responds; the first is a small Llama 3.1 instruct model (the same family as the reference study). API keys are taken from the notebook's secret store or a hidden prompt and are not saved to disk. The wrapper tries each call up to five times. If an error occurs and the message

indicates rate limiting, it waits 8, 16, 24 seconds and so on; for any other error it waits 2 seconds. On the final attempt an error message is printed and an empty response is returned, so that the caller can decide what to do. One of the models spends part of its token limit on reasoning, so that model is given a higher token limit to ensure it does not return an empty response.

7.4 Loading AltDiffusion

The AltDiffusion repository declares a custom text-encoder module that current diffusers releases cannot resolve. The loader therefore downloads the repository with a resumable snapshot call (skipping checkpoint formats and images that are not needed), rewrites the offending entry in the model index file, imports the encoder class that ships with diffusers under either of two module paths, loads the text encoder explicitly and passes it to the pipeline. The safety checker is disabled to save memory. If the class or the pipeline cannot be imported, the error message tells the user to switch to SDXL. Before a pipeline loads, any previously loaded pipeline and any evaluator are released and the GPU cache is emptied, since a T4 cannot hold a generator and an evaluator at once.

7.5 Checkpointing Primitives

Four small functions give the pipeline its durability: an atomic text writer, an atomic image writer, an append-only JSON-lines writer and a tolerant JSON-lines reader. Listing 1 shows the image writer and the append function.

```
def _atomic_save_image(img, path):
    path = Path(path)
    tmp = path.with_name(path.stem + ".tmp.png")
    img.save(tmp)
    os.replace(tmp, path)    # atomic rename


def append_jsonl(path, obj):
    with open(path, "a", encoding="utf-8") as f:
        f.write(json.dumps(obj, ensure_ascii=False) + "\n")
        f.flush()
        os.fsync(f.fileno())    # survive a sudden kill
```

Listing 1: Atomic image write and durable append

The reader skips blank or malformed lines, so a line truncated by a crash does not stop the next run, and later lines update earlier ones with the same key. This makes it possible to add the aesthetic score to a record long after its CLIP score was stored.

7.6 Caption Cache and the Fallback Rule

Captions are cached under a key made of the strategy, nationality, age, gender and landmark, with a language suffix for translations. When the multi-agent strategy is requested without a working language-model client, or when the conversation log shows that a fallback was used, the caption function raises an error instead of storing a caption (Listing 2). The effect is that the cache contains only genuine multi-agent captions.

```
cap, log = mosaig_caption(nat, age, gen, lm)
if is_fallback(log):
    raise RuntimeError("LLM call failed; progress is saved,"
                       " re-run this cell later")
cache[key] = cap
_save_cache(cache)
```

Listing 2: Refusing to cache a degraded caption

7.7 Scoring Implementation

CLIP similarity is computed with the ViT-L/14 model in half precision on the GPU. Image and text embeddings are normalised, and their dot product is the score. The knowledge score replaces the landmark in the caption by a landmark from a different country and subtracts the new similarity from the original. The replacement is selected by hashing the landmark name together with a salt, so the choice is reproducible but still varies across cases. The Inception Score uses torchvision's Inception-v3 on images resized to 299 × 299 and averages over five splits. The aesthetic predictor runs on the CPU to leave GPU memory for the generators; it is loaded only once and invoked in chunks of six images. When the predictor cannot be installed, the aesthetic column stays empty instead of stopping the run.

7.8 Studio Interface

The interface has three tabs. The first accepts a free-text prompt in any of the five languages and offers choices of model, strategy (multi-agent, simple or raw), number of candidates from one to six, preset, seed, whether to use aesthetics in ranking, and whether AltDiffusion should receive the user's own language. It shows the best image, the ranked gallery, the captions, the score table and the agent log. The second tab offers the same pipeline through drop-down lists for nationality, age, gender, landmark and language. The third renders one simple and one multi-agent image for the same fields and seed side by side, together with their CLIP scores, which is the quickest way to see the effect of caption composition. Errors raised inside the pipeline are surfaced as readable messages in the interface. Each request creates a folder named by its time stamp that contains every candidate, BEST.png and metadata.json. The ranking blend is shown in Listing 3.

```
parts = {"clip": 0.45, "knowledge": 0.30, "aesthetic": 0.25}
score, tot = np.zeros(len(df)), 0
for c, w in parts.items():
    if df[c].notna().any():
        score += w * minmax(df[c]).fillna(0.5)
        tot += w
df["final_score"] = score / max(tot, 1e-9)
df = df.sort_values("final_score", ascending=False)
```

Listing 3: Weighted blend used to rank candidates

7.9 Reproducibility Features

Table 7 shows how the implementation differs from a normal experimental script. These properties ensure that experiments can be repeated many times in an unreliable environment.

Table 7: Reproducibility and robustness features

Feature	Behaviour
Fixed shuffled order	Cases are drawn from a shuffled grid with a fixed seed, so enlarging N keeps all earlier cases
Same seed per case	The simple and multi-agent images of case i share the noise seed s0 + i
Atomic image writes	Temporary file then rename; unreadable files are regenerated
Append-only score log	One line per score, flushed to disk; tolerant reader
Caption hash in score key	Changing a caption invalidates its old score
No cached fallbacks	Degraded captions are never stored or scored as multi-agent
Pinned package versions	Avoids silent API changes in generator pipelines
Blinded sheet protection	Existing rating sheets are never overwritten

8. RESULT DISCUSSION AND COMPARATIVE ANALYSIS

8.1 Experimental Setup

Table 8 records the configuration of the run that produced the results below. The default notebook settings are a twelve-case benchmark with both strategies in English, four base cases for the fairness probe and a 24-image rating sample; larger values reuse everything already stored.

Table 8: Configuration of the evaluated run

Parameter	Value
Image model	AltDiffusion-m18 / SDXL-base
Preset	fast / paper
Benchmark cases N	12 by default
Strategies	Simple template and multi-agent (two rounds)
Languages evaluated	English only
Fairness base cases	4 by default
Seed	42 for the grid shuffle; per-case generator seed = model seed + case index
GPU / session	NVIDIA T4

8.2 Simple Versus Multi-Agent Captions

Table 9 shows the mean results for each metric. The aesthetic scores have been divided by 10, and the Inception Score divided by the number of images to give a maximum value of 1. This was done to make the results easier to plot, and it differs from other studies. The Inception column should therefore be compared between strategies within this run only.

Table 9: Mean scores per caption strategy

Scores

clip	knowledge	wrong_landmark	cand	aesthetic	final_score
0.353	0.048	Meridian Gate of Huế	2	4.293	0.75
0.291	0.001	Meridian Gate of Huế	3	5.112	0.292
0.285	0.003	Meridian Gate of Huế	1	4.986	0.221

8.3 Comparison with the Reference Study

The original paper states the average results of its simple and multi-agent English captions on a much larger set of images [1]. The values given in the text of that paper are shown in Table 10. These results are not directly comparable with ours, as there are differences in the size of

the sample, the language-model service used, the aesthetic and Inception normalisation, and the generator parameters. However, the table shows whether the changes have the same effect.

Table 10: Direction of change from simple to multi-agent captions

Measure	AltDiffusion (English)	FLUX (English)
Quality ↑	0.48 → 0.77	0.45 → 0.65
Knowledge ↑	0.42 → 0.52	0.43 → 0.57
Fairness ↓	0.32 → 0.17	0.34 → 0.22
Alignment	Comparable	Comparable
Aesthetics	Largely unchanged	Largely unchanged

Agreement in direction on quality, knowledge and fairness would support the claim that decomposed captions help, even at a smaller scale and with a different language-model back end.

9. CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

A Multi-Agent Framework was designed to test whether language-model agents, by role-playing, can produce better captions for cross-cultural scenes on low-powered hardware. The framework contains a moderator, three persona agents and a summarizer. The caption governor keeps captions within the 77-token limit and protects the landmark. Images were created using AltDiffusion, SDXL or FLUX. The framework supports human analysis of blinded images using the weighted kappa statistic, and a studio that ranks candidates by a weighted sum of three normalised signals.

Care was taken with the engineering and the prompts. Atomic writes were used and scores were recorded append-only. Captions were hashed into the score key, and no fallback captions were cached, to ensure an honest experiment. This means that any interruption does not affect accuracy. Due to the small sample size, the proxy metrics and the limited human study, the results should be seen as indicative.

9.2 Future Scope

1. Scale up the baseline to the full 750-prompt grid and five languages, reusing the stored captions through the resumable runner. Perform an ablation to show the effect of multi-agent decomposition, and compare against a single-agent, template-based caption of equal length.
2. Replace CLIP-based scoring with attribute detectors, landmark classifiers and question-answering models that extract the composition and identity attributes of the monuments. Expand the demographics further; for example, nationality can be broken down into more specific regional or community categories, with advice taken from individuals in those communities.
3. Add activities, festivals and everyday scenes to the landmark setting, and test whether the same caption pipeline helps there.
4. Compare several language models as agents, and study whether a small local model is sufficient, which would remove the dependency on a hosted service.
5. Learn the ranking weights of the studio from human preferences instead of fixing them by hand.

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LIST OF PUBLICATIONS

[To be added by the students, if any.]

PLAGIARISM REPORT (TURNITIN)

[Attach the Turnitin similarity report here.]